

Tutorial 6

CS5487 Lecture Notes (2024A)
Prof. Antoni B. Chan
Dept of Computer Science
City University of Hong Kong

PS6.6: Gaussian Classifier

prior: $p(y=j) = \pi_j$, $j \in \{1, \dots, K\}$ shared cov for all classes.

CCD: $p(x|y=j) = N(x|\mu_j, \Sigma_j)$

different mean for each class

BDR w/ 0-1 loss:

$$g(x) = \operatorname{argmax}_j \underbrace{\log p(x|y=j) + \log p(y=j)}_{g_j(x) = \text{discriminant function for class } j.}$$

$$\begin{aligned} a) \quad g_j(x) &= \log p(x|j) + \log p(y=j) \\ &= \underbrace{-\frac{1}{2} \|x - \mu_j\|_2^2}_{\text{constant w.r.t } j \text{ (drop them)}} - \underbrace{\frac{1}{2} \log |\Sigma| - \frac{d}{2} \log 2\pi}_{\text{constant w.r.t } j} + \log \pi_j \\ &= -\frac{1}{2} (x - \mu_j)^T \Sigma^{-1} (x - \mu_j) + \log \pi_j + \text{const} \\ &= -\frac{1}{2} \left(\underbrace{x^T \Sigma^{-1} x}_{\text{const. w.r.t } j} - 2x^T \Sigma^{-1} \mu_j + \mu_j^T \Sigma^{-1} \mu_j \right) + \log \pi_j + \text{const} \\ &= \underbrace{x^T \Sigma^{-1} \mu_j}_{w_j} - \underbrace{\frac{1}{2} \mu_j^T \Sigma^{-1} \mu_j}_{b_j} + \log \pi_j + \text{const.} \end{aligned}$$

$$\Rightarrow g_j(x) = w_j^T x + b_j, \quad w_j = \Sigma^{-1} \mu_j, \quad b_j = -\frac{1}{2} \mu_j^T \Sigma^{-1} \mu_j + \log \pi_j$$

(b) consider classes (i, j) , $i \neq j$

The decision btwn i & j is when $g_i(x) = g_j(x)$
i.e. can't tell if it's class i or class j .

$$g_i(x) = g_j(x)$$

$$w_i^T x + b_i = w_j^T x + b_j$$

$$w_i^T x - w_j^T x + b_i - b_j = 0$$

$$\underbrace{(w_i - w_j)^T x}_w + \underbrace{(b_i - b_j)}_b = 0 \Rightarrow w^T x + b = 0 \quad (\text{hyperplane})$$

$$w = w_i - w_j = \Sigma^{-1} \mu_i - \Sigma^{-1} \mu_j = \Sigma^{-1} (\mu_i - \mu_j)$$

$$\begin{aligned} b &= b_i - b_j = -\frac{1}{2} \mu_i^T \Sigma^{-1} \mu_i + \log \pi_i + \frac{1}{2} \mu_j^T \Sigma^{-1} \mu_j - \log \pi_j \\ &= -\frac{1}{2} (\mu_i^T \Sigma^{-1} \mu_i - \mu_j^T \Sigma^{-1} \mu_j) + \log \frac{\pi_i}{\pi_j} \end{aligned}$$

$$(a-b)^T(a+b) = a^T a - b^T b$$

$$a^2 - b^2 = (a-b)(a+b)$$

$$a^T b - b^T a = 0$$

$$b = -\frac{1}{2} (\mu_i + \mu_j)^T \Sigma^{-1} (\mu_i - \mu_j) + \log \frac{\pi_i}{\pi_j}$$

$$x) \Rightarrow w^T x + b = 0$$

$$\Rightarrow w^T (x - x_0) = 0$$

$$w^T x - w^T x_0 = 0$$

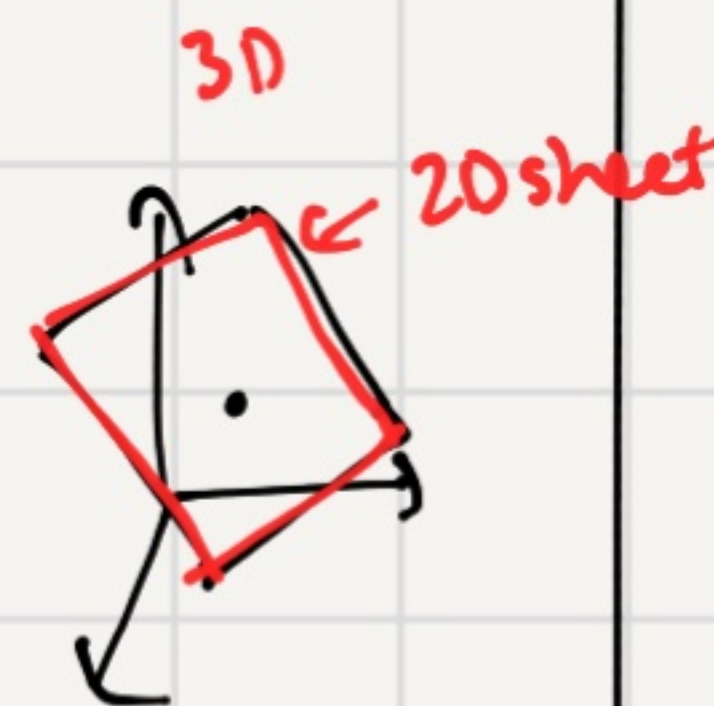
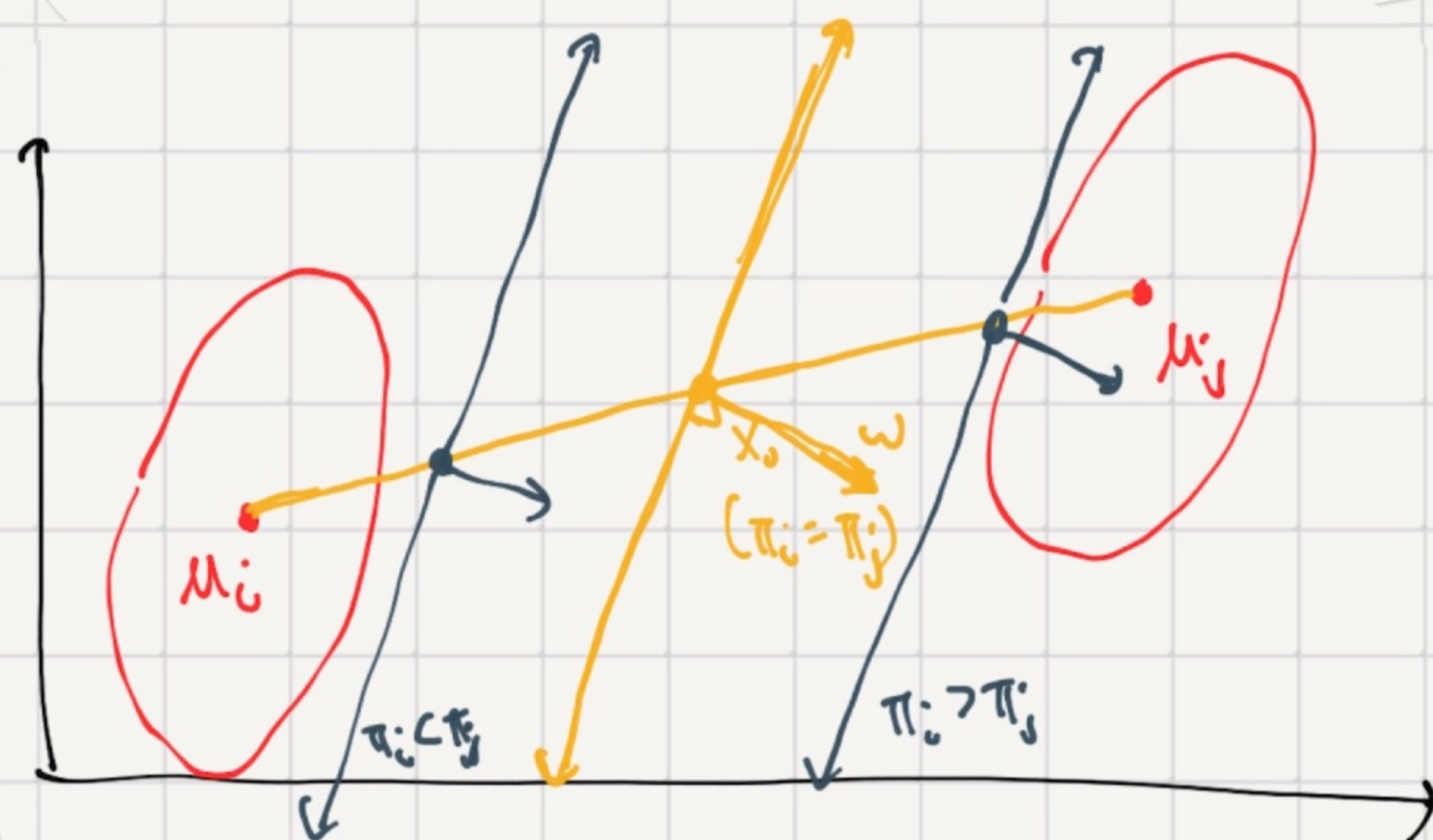
$$b = -w^T x_0$$

$$-\frac{1}{2} (\mu_i + \mu_j)^T \Sigma_i^{-1} (\mu_i - \mu_j) + \log \frac{\pi_i}{\pi_j} = -(\mu_i - \mu_j)^T \Sigma_i^{-1} x_0$$

$$= \frac{\|\mu_i - \mu_j\|_{\Sigma_i}^2}{\|\mu_i - \mu_j\|_{\Sigma_i}^2} = \frac{(\mu_i - \mu_j)^T \Sigma_i^{-1} (\mu_i - \mu_j)}{\|\mu_i - \mu_j\|_{\Sigma_i}^2}$$

$$-(\mu_i - \mu_j)^T \Sigma_i^{-1} \left[\frac{\mu_i + \mu_j}{2} - \frac{(\mu_i - \mu_j)}{\|\mu_i - \mu_j\|_{\Sigma_i}^2} \log \frac{\pi_i}{\pi_j} \right] = -(\mu_i - \mu_j)^T \Sigma_i^{-1} x_0$$

$$\therefore x_0 = \frac{\mu_i + \mu_j}{2} - \frac{(\mu_i - \mu_j)}{\|\mu_i - \mu_j\|_{\Sigma_i}^2} \log \frac{\pi_i}{\pi_j}$$



PS 6.2 BDR & Regression

output $y \in \mathbb{R}$

input $x \in \mathbb{R}^d$

Suppose we have $p(y|x)$

a) Select the optimal y^* for a given x .

$$L(g(x), y) = (g(x) - y)^2$$

$$\text{conditional risk: } R(x) = E_{y|x} [L(g(x), y)]$$

$$= \int p(y|x) (g(x) - y)^2 dy$$

$$= \int p(y|x) (g^2(x) - 2g(x)y + y^2) dy$$

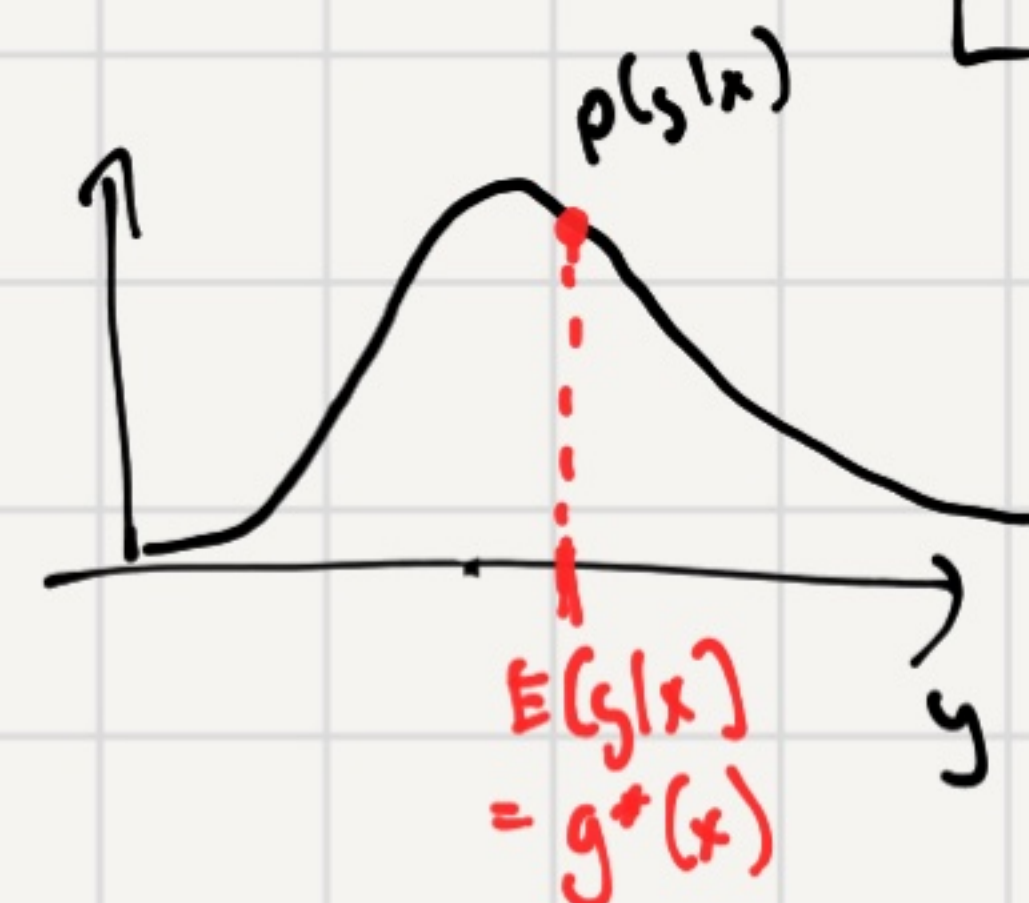
$$\int p(y|x) y dy = g(x) \int p(y|x) dy = g(x) E[y|x]$$

$$R(x) = g^2(x) - 2g(x)E[y|x] + E[y^2|x]$$

minimize $R(x)$ wrt. $g(x)$:

$$\frac{\partial R(x)}{\partial g(x)} = 2g(x) - 2E[y|x] = 0$$

$$g^*(x) = E[y|x]$$



$$L(g(x), y) = |g(x) - y|$$

$$\Rightarrow g^*(x) = \text{median } p(y|x)$$